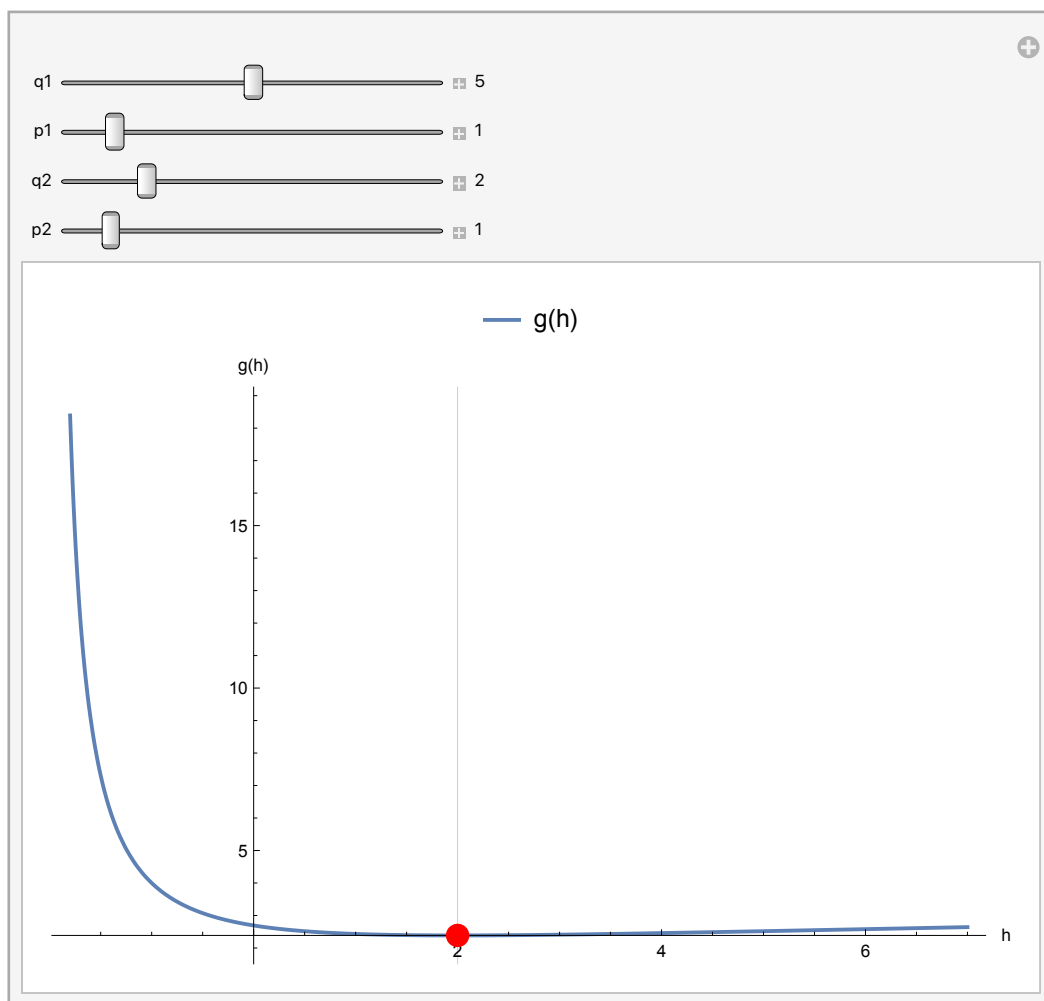


```

In[*]:= Manipulate[Module[{g, hinf, hmin, hmax, showDot, dotCoord},
  g[h_] := Abs[q1 - p1] / (q2 + h) + Log[(q2 + h) / p2];
  hinf = Abs[q1 - p1] - q2;
  hmin = Max[-0.9 q2, -q2 + 0.01];
  (*keep domain valid*)hmax = hinf + 5;
  showDot = hmin < hinf < hmax && Quiet[NumericQ[g[hinf]]] &&
    Im[g[hinf]] == 0 && Re[g[hinf]] < 10^6 && Re[g[hinf]] > -10^6;
  (*avoid wild values*)dotCoord = {hinf, g[hinf]};
  Plot[g[h], {h, hmin, hmax}, PlotRange -> All,
    AxesLabel -> {"h", "g(h)"}, Epilog -> If[showDot, {Red, PointSize[0.025],
      Point[dotCoord], Text[Style["", Red, 12], dotCoord, {0, 1}]}], {}],
    PlotLegends -> Placed[{"g(h)"}, Above], GridLines -> {{hinf}, None},
    ImageSize -> 500]], {{q1, 5}, 0, 10, Appearance -> "Labeled"},
  {{p1, 1}, 0, 10, Appearance -> "Labeled"},
  {{q2, 2}, 0.1, 10, Appearance -> "Labeled"},
  {{p2, 1}, 0.1, 10, Appearance -> "Labeled"},
  ControlPlacement -> Top]

```

Out[*]=



In[*]:=

```

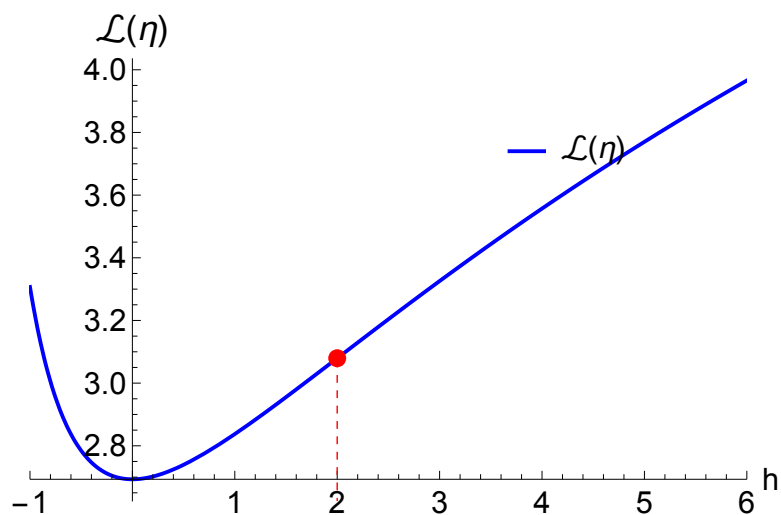
In[*]:= (*Parameters*) q1 = 4; p1 = 0; q2 = 2; p2 = 1;
d = Abs[q1 - p1];
hinfl = d - q2;

g[h_] := Log[(q2 + h)^2 / (q2 * p2)] + d / (q2 + h);

Plot[g[h], {h, -1, 6}, PlotStyle -> {Blue, Thick},
  AxesLabel -> {"h", Style[" $\mathcal{L}(\eta)$ ", 18]}],
  PlotLegends -> Placed[{" $\mathcal{L}(\eta)$ "}, {0.75, 0.8}],
  Epilog -> {{Dashed, Red, Line[{hinfl, 0}, {hinfl, g[hinfl]}]},
    Red, PointSize[0.025], Point[{hinfl, g[hinfl]}]},
    Text[Style["h_{inf}", Red, 16], {hinfl, -2}, {0, 1}]}},
  PlotRange -> {{-1, 6}, All}, TicksStyle -> Directive[Black, 14],
  LabelStyle -> Directive[Black, 16], ImageSize -> 400]

```

Out[*]=

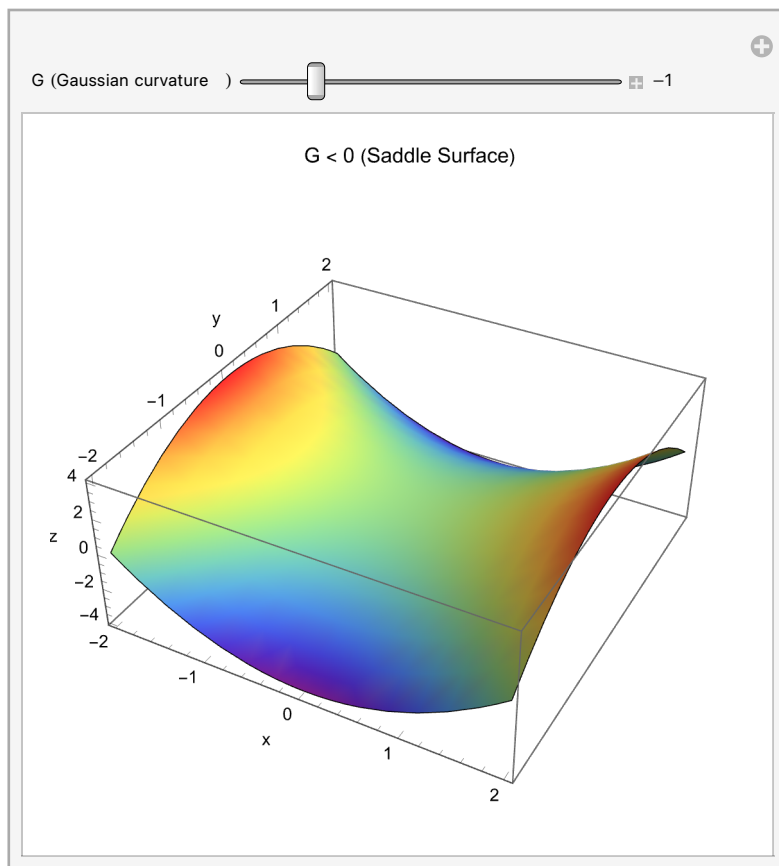


```

In[*]:= Manipulate[Module[{surf}, surf =
  Which[G < -0.01, (*Saddle:negative curvature*)Plot3D[x^2 + G y^2, {x, -2, 2},
    {y, -2, 2}, Mesh → None, PlotRange → All, ColorFunction → "Rainbow",
    AxesLabel → {"x", "y", "z"}, PlotLabel → "G < 0 (Saddle Surface)", G > 0.01,
    (*Elliptic:positive curvature (sphere patch)*)ParametricPlot3D[
    {Sin[u] × Cos[v], Sin[u] × Sin[v], Cos[u] / Sqrt[G]}, {u, 0, Pi},
    {v, 0, 2 Pi}, Mesh → None, PlotRange → All, ColorFunction → "Rainbow",
    AxesLabel → {"x", "y", "z"}, PlotLabel → "G > 0 (Spherical Patch)", True,
    (*G≈0:Cylinder*)ParametricPlot3D[{Cos[v], y, Sin[v]}, {y, -2, 2},
    {v, 0, 2 Pi}, Mesh → None, PlotRange → All, ColorFunction → "Rainbow",
    AxesLabel → {"x", "y", "z"}, PlotLabel → "G = 0 (Cylinder)"]];
surf], {{G, -1, "G (Gaussian curvature)"}, -1.5, 1.5, Appearance → "Labeled"]}

```

Out[*]=



```

In[*]:= (*Hyperbolic distance in Poincare disk:*)
hyperDist[u_, v_] := ArcCosh[1 + 2 Norm[u - v]^2 / ((1 - Norm[u]^2) (1 - Norm[v]^2))];

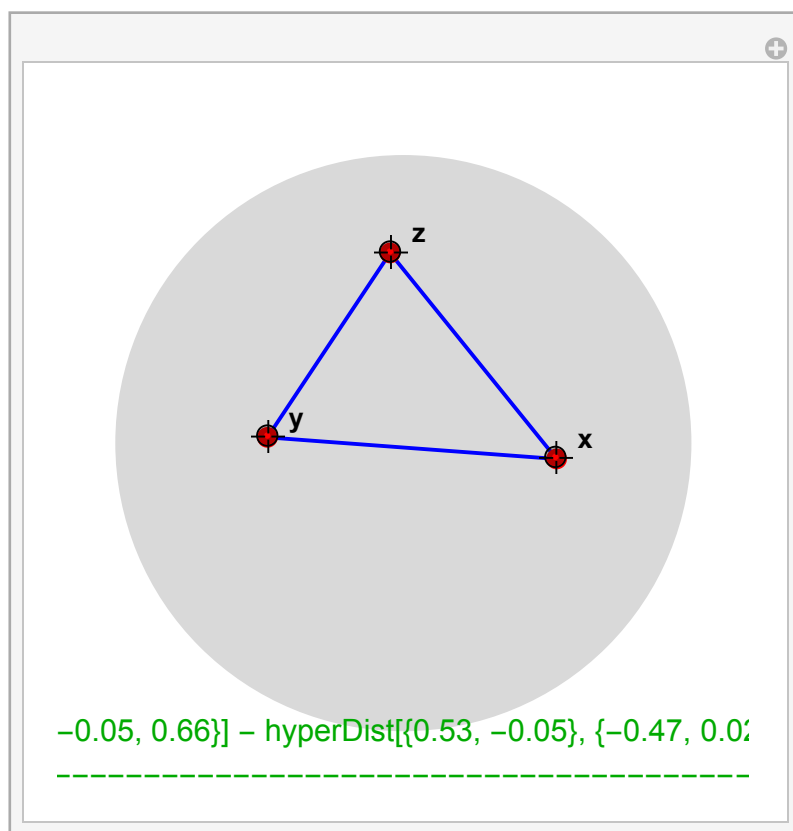
```

```

Manipulate[Module[{pts = {x, y, z}, gromov},
  gromov = 1 / 2 (hyperDist[x, z] + hyperDist[y, z] - hyperDist[x, y]);
  Graphics[{LightGray, Disk[{0, 0}, 1], Blue, Thick, Line[{x, y}],
    Line[{y, z}], Line[{z, x}], Red, PointSize[0.03], Point[x], Point[y],
    Point[z], Black, Text[Style["x", 14, Bold], x + {0.1, 0.07}],
    Text[Style["y", 14, Bold], y + {0.1, 0.07}],
    Text[Style["z", 14, Bold], z + {0.1, 0.07}], Text[Style["Gromov product: "<>
    ToString@NumberForm[gromov, {3, 2}], 16, Darker[Green]], {0, -1.15}]],
  PlotRange -> {{-1.2, 1.2}, {-1.2, 1.2}}, Axes -> False]],
{{x, {0.7, 0}}, Locator, LocatorRegion -> Disk[]},
{{y, {-0.7, 0}}, Locator, LocatorRegion -> Disk[]},
{{z, {0, 0.7}}, Locator, LocatorRegion -> Disk[]},
TrackedSymbols -> {x, y, z}]

```

Out[*]=



```

In[*]:= myDist[p_, q_] := Module[{m, dx}, dx = Abs[p[[1]] - q[[1]]];
  m = Max[p[[2]], q[[2]], dx / 2];
  Log[m / p[[2]]] + Log[m / q[[2]]] + 1 / m * Log[dx]];

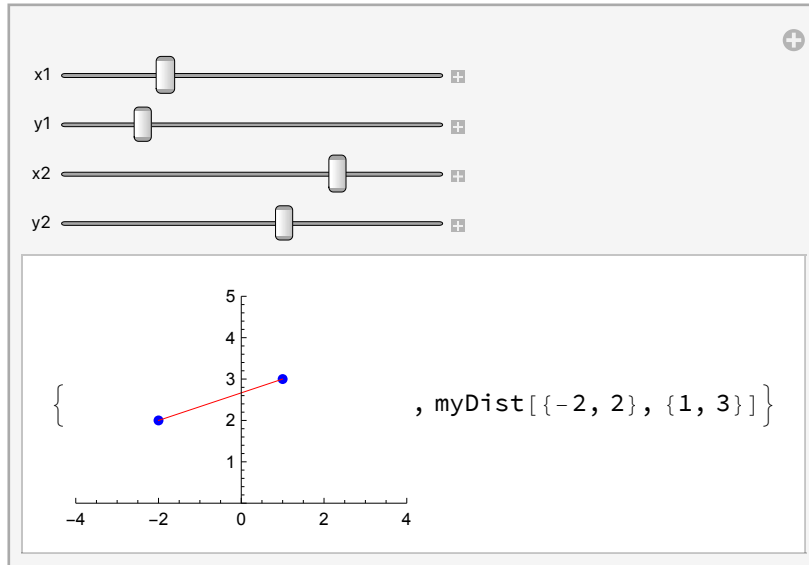
```

```

Manipulate[
  {Graphics[{Blue, PointSize[0.03], Point[{x1, x2}], Point[{y1, y2}], Red,
    Line[{x1, x2}, {y1, y2}]}], PlotRange -> {{-4, 4}, {0, 5}}, Axes -> True],
  myDist[{x1, x2}, {y1, y2}], {{x1, -2}, -4, 4},
  {{y1, 1}, 0.1, 5}, {{x2, 2}, -4, 4}, {{y2, 3}, 0.1, 5]}

```

Out[*]=



```

In[*]:= myDist[p_, q_] := Log[Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] / p[[2]]] +
  Log[Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] / q[[2]]] +
  1 / Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] * Abs[p[[1]] - q[[1]]];

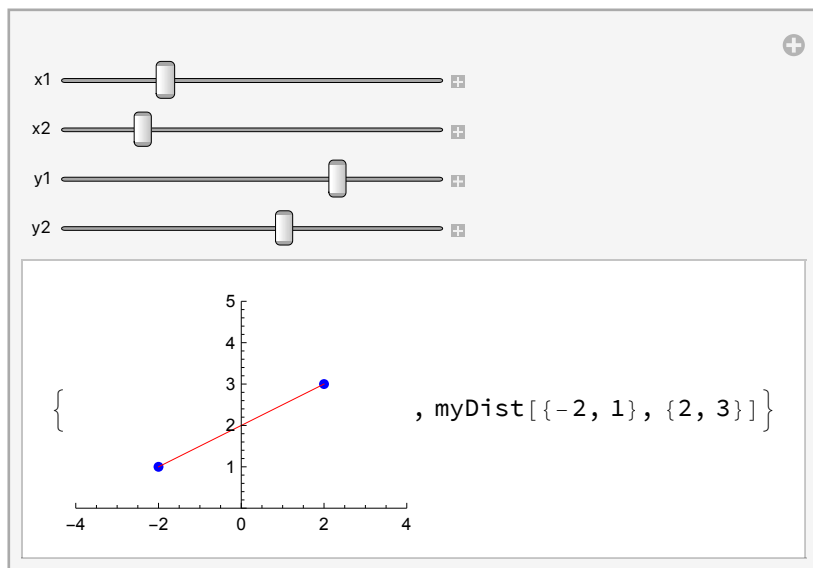
```

```

Manipulate[
  {Graphics[{Blue, PointSize[0.03], Point[{x1, x2}], Point[{y1, y2}], Red,
    Line[{x1, x2}, {y1, y2}]}], PlotRange -> {{-4, 4}, {0, 5}}, Axes -> True],
  myDist[{x1, x2}, {y1, y2}], {{x1, -2}, -4, 4},
  {{x2, 1}, 0.1, 5}, {{y1, 2}, -4, 4}, {{y2, 3}, 0.1, 5]}

```

Out[*]=



```

In[*]:= myDist[p_, q_] := Log[Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] / p[[2]]] +
  Log[Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] / q[[2]]] +
  1 / Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] * Abs[p[[1]] - q[[1]]];

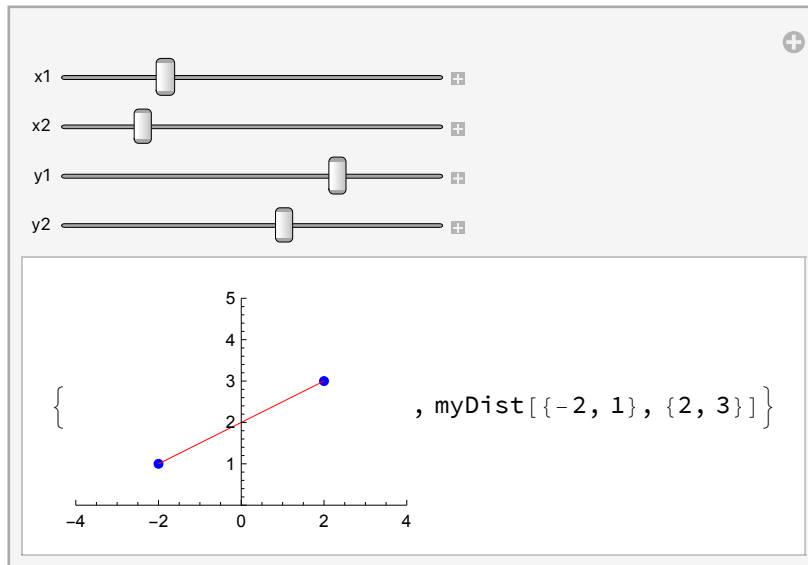
```

```

Manipulate[
  {Graphics[{Blue, PointSize[0.03], Point[{x1, x2}], Point[{y1, y2}], Red,
    Line[{x1, x2}, {y1, y2}]}], PlotRange -> {{-4, 4}, {0, 5}}, Axes -> True],
  myDist[{x1, x2}, {y1, y2}], {{x1, -2}, -4, 4},
  {{x2, 1}, 0.1, 5}, {{y1, 2}, -4, 4}, {{y2, 3}, 0.1, 5]}

```

Out[*]=



```

In[*]:= myDist[p_, q_] := Log[Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] / p[[2]]] +
  Log[Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] / q[[2]]] +
  1 / Max[p[[2]], q[[2]], Abs[p[[1]] - q[[1]] / 2] * Abs[p[[1]] - q[[1]]];

```

```

Γ[x_, y_, z_] := (myDist[x, z] + myDist[y, z] - myDist[x, y]) / 2;

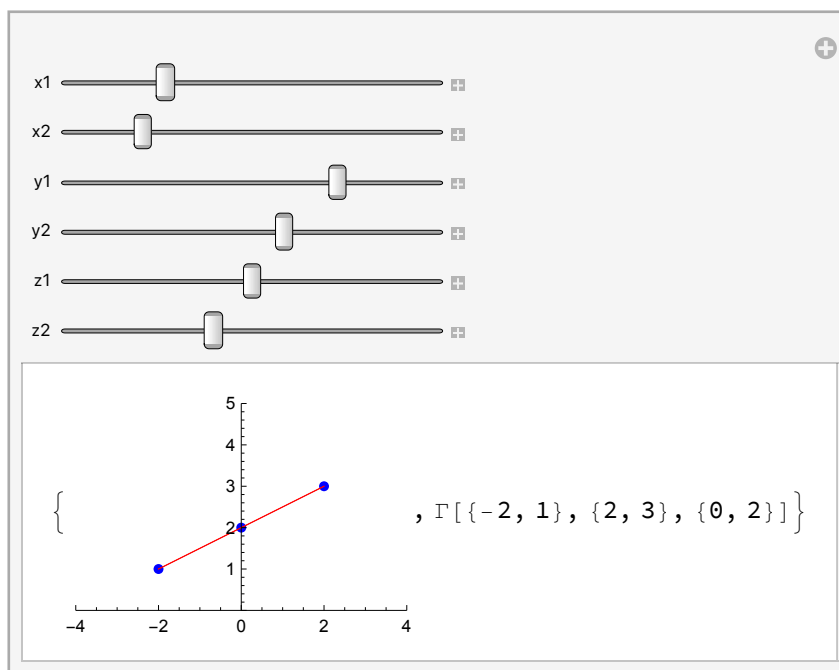
```

```

Manipulate[Graphics[{Blue, PointSize[0.03], Point[{x1, x2}], Point[{y1, y2}],
  Point[{z1, z2}], Red, Line[{x1, x2}, {y1, y2}], Line[{x1, x2}, {z1, z2}],
  Line[{y1, y2}, {z1, z2}], PlotRange → {{-4, 4}, {0, 5}}, Axes → True],
  Γ[{x1, x2}, {y1, y2}, {z1, z2}], {{x1, -2}, -4, 4},
  {{x2, 1}, 0.1, 5}, {{y1, 2}, -4, 4}, {{y2, 3}, 0.1, 5},
  {{z1, 0}, -4, 4}, {{z2, 2}, 0.1, 5}]

```

Out[*]=



```

In[*]:= 3.4 * 1000

```

Out[*]=

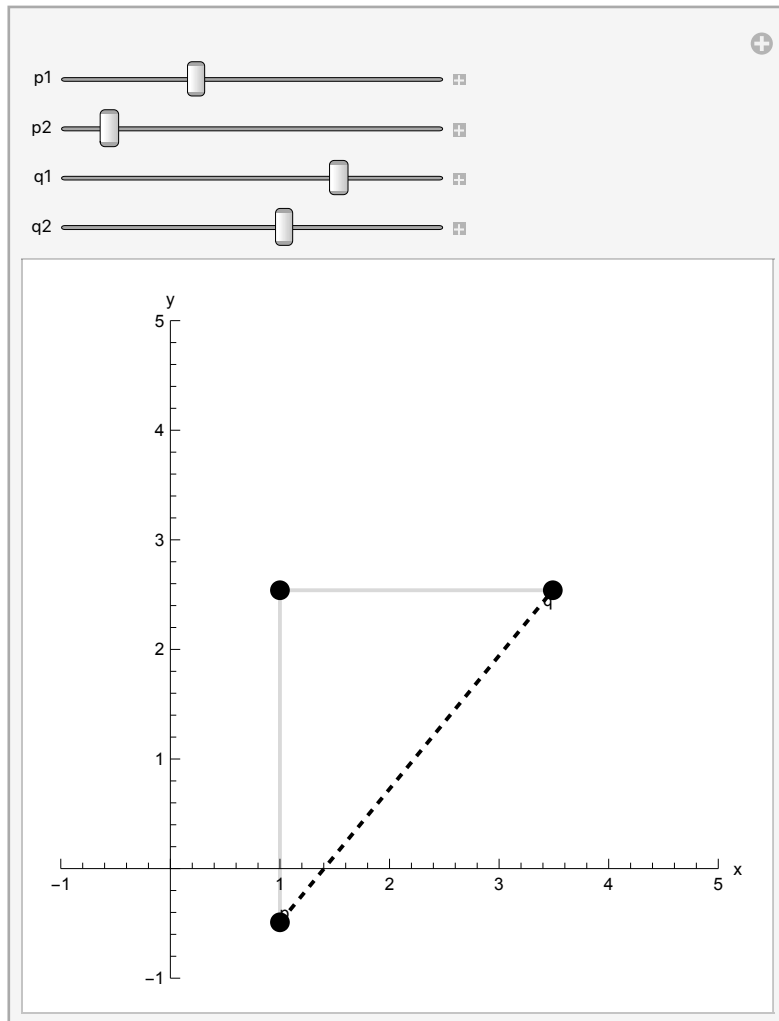
3400.

```

In[*]:= Manipulate[Graphics[
  {LightGray, Thick, Line[{{p1, p2}, {p1, q2}}], Line[{{p1, q2}, {q1, q2}}],
  Black, PointSize[0.03], Point[{p1, p2}], Point[{q1, q2}], Point[{p1, q2}],
  Black, Dashed, Line[{{p1, p2}, {q1, q2}}]}, PlotRange → {{-1, 5}, {-1, 5}},
  Axes → True, AspectRatio → 1, AxesLabel → {"x", "y"},
  Epilog → {Text["p", {p1, p2}, {-1, -1}], Text["q", {q1, q2}, {1, 1}]}],
  {{p1, 1}, -1, 5}, {{p2, 1}, -1, 5}, {{q1, 4}, -1, 5}, {{q2, 3}, -1, 5}]

```

Out[*]:=



```

In[*]:= L[{p1_, p2_}, {q1_, q2_}] := Module[{m}, m = Max[p2, q2,  $\frac{\text{Abs}[q1 - p1]}{2}$ ];
  Log[ $\frac{m}{p2}$ ] + Log[ $\frac{m}{q2}$ ] +  $\frac{\text{Abs}[q1 - p1]}{m}$ ]

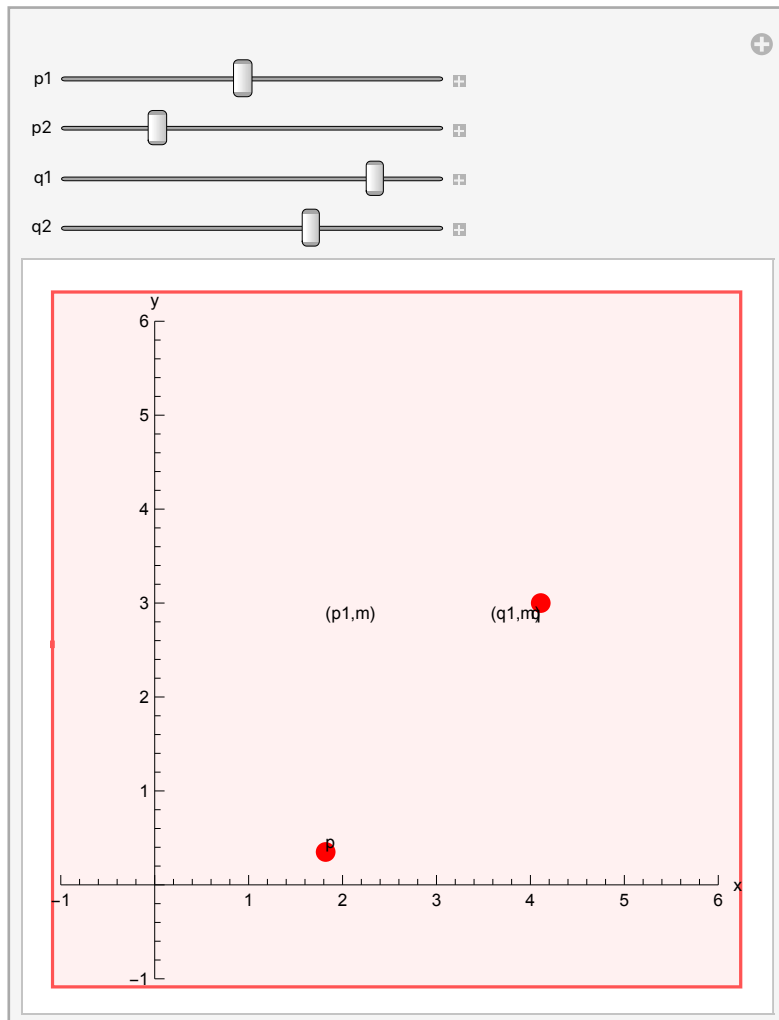
```

```

In[*]:= LPath[p_, q_] := Module[{m, pts}, m = Max[p[[2]], q[[2]], Abs[q[[1]] - p[[1]] / 2];
  pts = {p, {p[[1]], m}, {q[[1]], m}, q};
  (*Remove consecutive duplicates*)pts = DeleteDuplicates[pts];
  pts]
Manipulate[Graphics[{LightGray, Thick, Line[LPath[{p1, p2}, {q1, q2}]]},
  Red, PointSize[0.03], Point[{p1, p2}], Point[{q1, q2}], Black,
  PointSize[0.02], Point /@ Rest[Most[LPath[{p1, p2}, {q1, q2}]]],
  (*Bends only*)Text["p", {p1, p2}, {-1, -1}], Text["q", {q1, q2}, {1, 1}],
  Text["(p1,m)", {p1, Max[p2, q2, Abs[q1 - p1] / 2]}, {-1, 1}],
  Text["(q1,m)", {q1, Max[p2, q2, Abs[q1 - p1] / 2]}, {1, 1}],
  PlotRange -> {{-1, 6}, {-1, 6}}, Axes -> True, AspectRatio -> 1,
  AxesLabel -> {"x", "y"}], {{p1, 1}, -1, 5},
  {{p2, 1}, -1, 5}, {{q1, 4}, -1, 5}, {{q2, 3}, -1, 5}]

```

Out[*] =



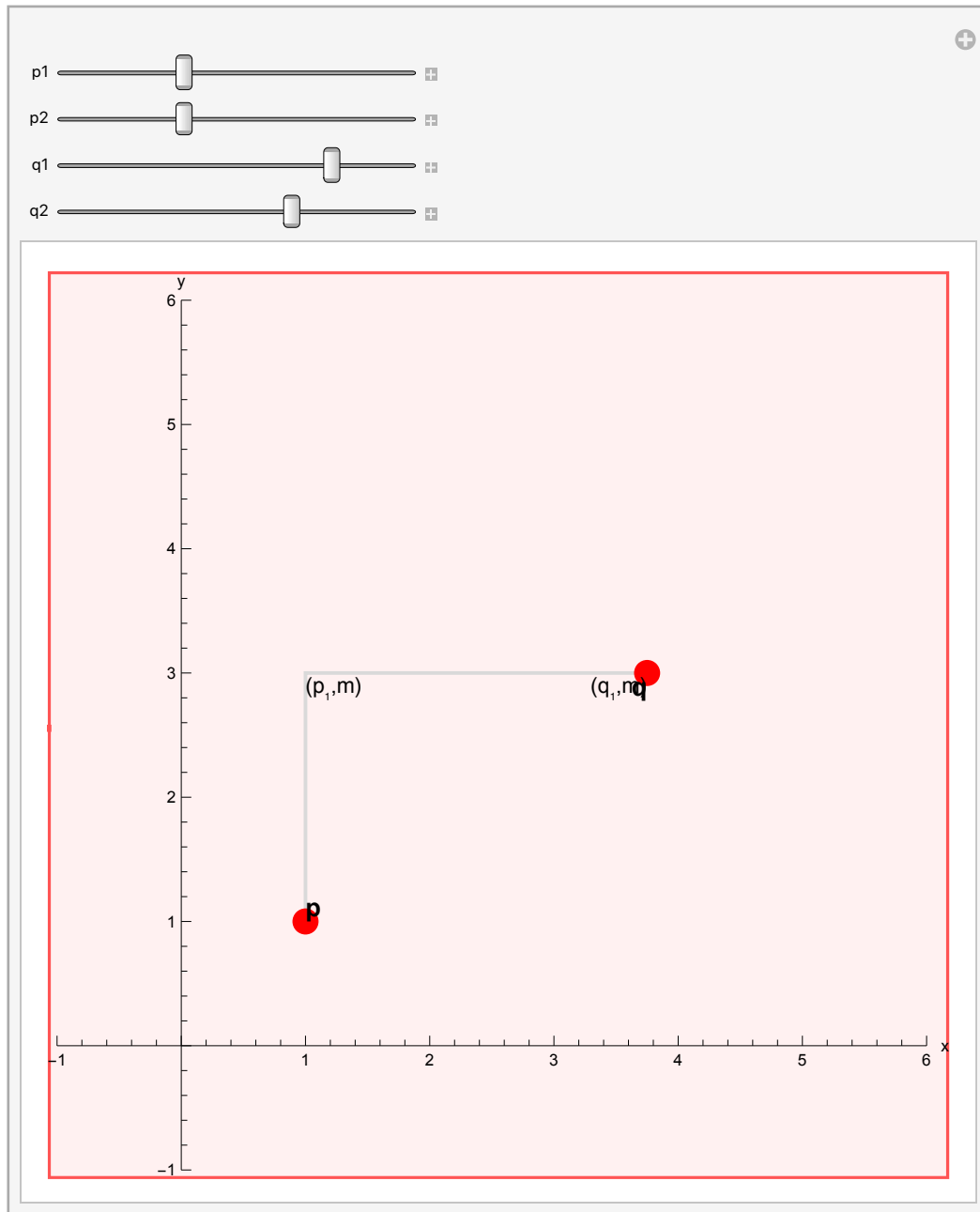
```

In[*]:= (*Define the path function*)
LPath[p_, q_] := Module[{m, pts}, m = Max[p[[2]], q[[2]], Abs[q[[1]] - p[[1]] / 2];
  pts = DeleteDuplicates[{p, {p[[1]], m}, {q[[1]], m}, q}];

Manipulate[Graphics[(*First Path:Vertical-Horizontal path*)
  {LightGray, Thick, Line[{p1, p2}, {p1, q2}, {q1, q2}]}],
(*Second Path:Taxicab path with bends*){Blue, Thick,
  Line[LPath[{p1, p2}, {q1, q2}]}], (*Points p,q and intermediate points*)
{Red, PointSize[0.03], Point[{p1, p2}], Point[{q1, q2}]},
{Black, PointSize[0.02], Point /@ Rest[Most[LPath[{p1, p2}, {q1, q2}]]]},
(*Labels*)Text[Style["p", Bold, 14], {p1, p2}, {-1, -1}],
Text[Style["q", Bold, 14], {q1, q2}, {1, 1}],
Text[Style["(p,m)", 12], {p1, Max[p2, q2, Abs[q1 - p1] / 2]}, {-1, 1}],
Text[Style["(q,m)", 12], {q1, Max[p2, q2, Abs[q1 - p1] / 2]}, {1, 1}],
PlotRange -> {{-1, 6}, {-1, 6}}, Axes -> True, AspectRatio -> 1,
AxesLabel -> {"x", "y"}, ImageSize -> 500], {{p1, 1}, -1, 5},
{{p2, 1}, -1, 5}, {{q1, 4}, -1, 5}, {{q2, 3}, -1, 5}]

```

Out[]=

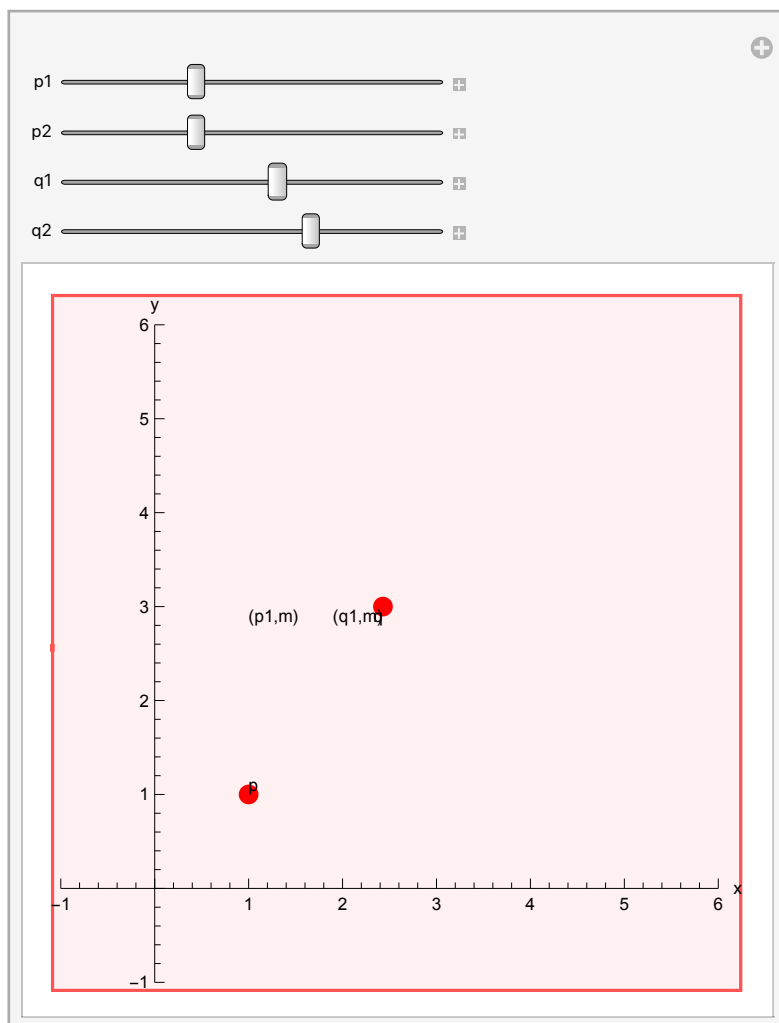


```

In[*]:= LPath[p_, q_] := Module[{m, pts}, m = Max[p[[2]], q[[2]], Abs[q[[1]] - p[[1]] / 2];
  pts = {p, {p[[1]], m}, {q[[1]], m}, q};
  (*Remove consecutive duplicates*)pts = DeleteDuplicates[pts];
  pts]
Manipulate[Graphics[{LightGray, Thick, Line[LPath[{p1, p2}, {q1, q2}]]},
  Red, PointSize[0.03], Point[{p1, p2}], Point[{q1, q2}], Black,
  PointSize[0.02], Point /@ Rest[Most[LPath[{p1, p2}, {q1, q2}]]],
  (*Bends only*)Text["p", {p1, p2}, {-1, -1}], Text["q", {q1, q2}, {1, 1}],
  Text["(p1,m)", {p1, Max[p2, q2, Abs[q1 - p1] / 2]}, {-1, 1}],
  Text["(q1,m)", {q1, Max[p2, q2, Abs[q1 - p1] / 2]}, {1, 1}],
  PlotRange -> {{-1, 6}, {-1, 6}}, Axes -> True, AspectRatio -> 1,
  AxesLabel -> {"x", "y"}], {{p1, 1}, -1, 5},
{{p2, 1}, -1, 5}, {{q1, 4}, -1, 5}, {{q2, 3}, -1, 5}]

```

Out[*]=



```
In[*]:=  $\mu[\{x1\_ , x2\_ \}, \{y1\_ , y2\_ \}] := \text{Max}[x2, y2, \text{Abs}[x1 - y1] / 2];$ 
```

```
a[\{x1\_ , x2\_ \}, \{y1\_ , y2\_ \}] := \{x1, \mu[\{x1, x2\}, \{y1, y2\}]\};
```

```
b[\{x1\_ , x2\_ \}, \{y1\_ , y2\_ \}] := \{y1, \mu[\{x1, x2\}, \{y1, y2\}]\};
```

```
geo[\{x1\_ , x2\_ \}, \{y1\_ , y2\_ \}] :=
```

```
Line[\{\{x1, x2\}, a[\{x1, x2\}, \{y1, y2\}], b[\{x1, x2\}, \{y1, y2\}], \{y1, y2\}\}\];
```

```
tri[\{x1\_ , x2\_ \}, \{y1\_ , y2\_ \}, \{z1\_ , z2\_ \}] :=
```

```
\{geo[\{x1, x2\}, \{y1, y2\}], geo[\{y1, y2\}, \{z1, z2\}], geo[\{z1, z2\}, \{x1, x2\}]\}
```

```
In[*]:= Graphics[geo[\{0, 1\}, \{3, 1\}]]
```

```
Out[*]=
```

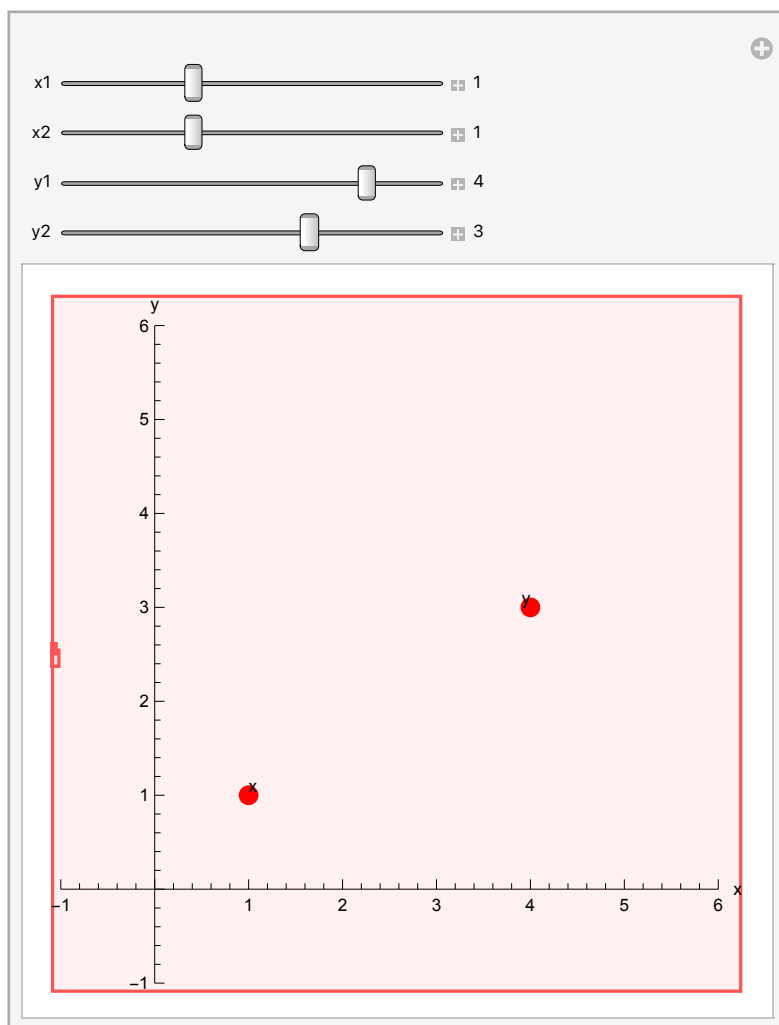


```

In[*]:= Manipulate[Graphics[{Blue, Thick, geo[{x1, x2}, {y1, y2}],
  (*Points*)Red, PointSize[0.03], Point[{x1, x2}], Point[{y1, y2}],
  Black, PointSize[0.02], Point[a[{x1, x2}, {y1, y2}]]],
  Point[b[{x1, x2}, {y1, y2}]]], (*Labels*)Text["x", {x1, x2}, {-1, -1}],
  Text["y", {y1, y2}, {1, -1}], Text["a", a[{x1, x2}, {y1, y2}], {-1, 1}],
  Text["b", b[{x1, x2}, {y1, y2}], {1, 1}]], PlotRange -> {{-1, 6}, {-1, 6}},
  Axes -> True, AspectRatio -> 1, AxesLabel -> {"x", "y"}],
  {{x1, 1}, -1, 5, Appearance -> "Labeled"},
  {{x2, 1}, -1, 5, Appearance -> "Labeled"},
  {{y1, 4}, -1, 5, Appearance -> "Labeled"},
  {{y2, 3}, -1, 5, Appearance -> "Labeled"}]

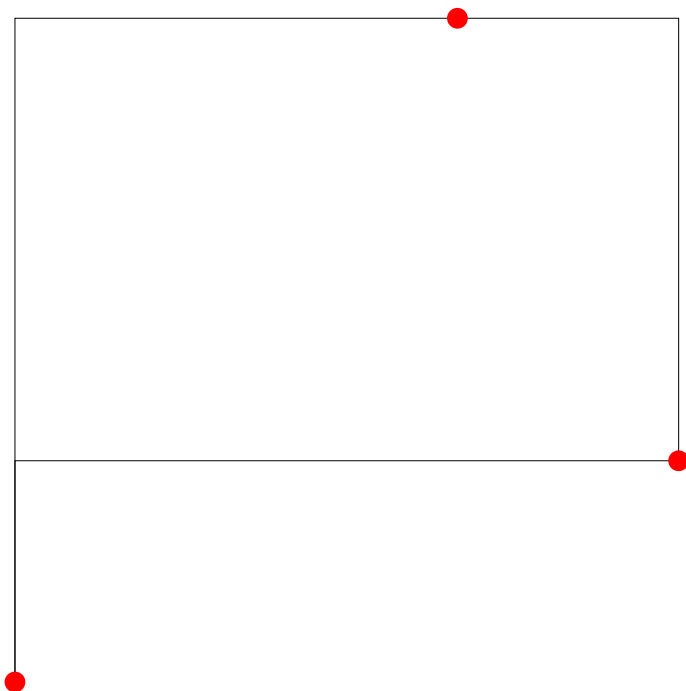
```

Out[*]=



```
In[*]:= Graphics[{tri[{0, 1}, {3, 2}, {2, 4}], Red,
  PointSize[0.03], Point[{0, 1}], Point[{3, 2}], Point[{2, 4}]}]
```

Out[*]=



```
In[*]:= N[x_] = x^2
```

... **N**: N[x_] does not contain a symbol to attach a rule to. [i](#)

Out[*]=

x^2

```
In[*]:= N[Pi]
```

Out[*]=

3.14159

```
In[*]:= ? N
```

Out[*]=

Symbol [i](#)

N[*expr*] gives the numerical value of *expr*.

N[*expr*, *n*] attempts to give a result with *n*-digit precision.



```
In[*]:= ? Line
```

Out[*]=

Symbol [i](#)

Line [{*p*₁, *p*₂, ...}] represents the line segments joining a sequence for points *p*_{*i*}.

Line [{*p*₁₁, *p*₁₂, ...}, {*p*₂₁, ...}, ...] represents a collection of lines.



```
(*
a) h(x,,z,w)
x,y,z,w are points of two coordinates
 $\mu(p,q) := \max\{p[[2]],q[[2]]\}$ 
```

```
*)
```

```
In[ ]:= ClearAll
```

```
Out[ ]=
```

```
ClearAll
```

```
In[ ]:=  $\mu[p_, q_] := \text{Max}[p[[2]], q[[2]]]$ 
```

```
In[ ]:=  $\nu[p_, q_] := \text{Min}[p[[1]], q[[1]]]$ 
```

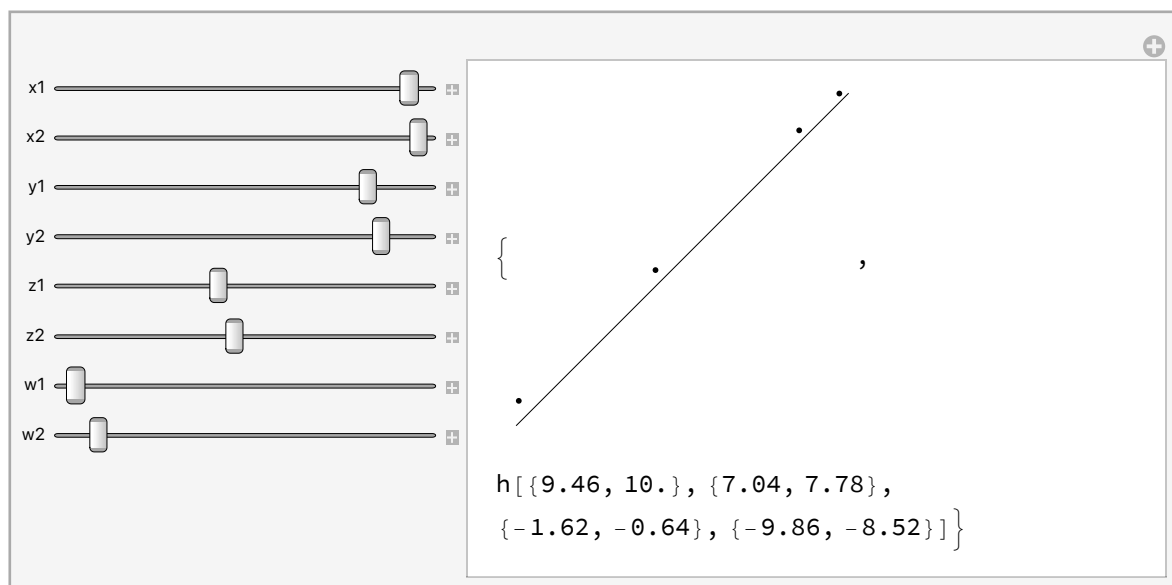
```
In[ ]:=  $h[x_, y_, z_, w_] := \text{Log}[e, ((\mu[x, y] - \nu[x, y]) * (\mu[z, w] - \nu[z, w])) / \text{Max}[(\mu[x, z] - \nu[x, z]) * (\mu[y, w] - \nu[y, w]), (\mu[x, w] - \nu[x, w]) * (\mu[z, y] - \nu[z, y])]]]$ 
```

```

In[ ]:= Manipulate[
  {
    Graphics[
      {
        Point[
          {
            {x1, x2}, {y1, y2}, {z1, z2}, {w1, w2}
          }
        ], Line[
          {
            {-10, -10}, {10, 10}
          }
        ]
      }
    ],
    h[{x1, x2}, {y1, y2}, {z1, z2}, {w1, w2}]
  },
  {{x1, 0}, -10, 10}, {{x2, 1}, -10, 10},
  {{y1, 0}, -10, 10}, {{y2, 1}, -10, 10},
  {{z1, 0}, -10, 10}, {{z2, 1}, -10, 10},
  {{w1, 0}, -10, 10}, {{w2, 1}, -10, 10}
]

```

Out[]=



```

In[ ]:= Log[E, 4]

```

Out[]=

Log[4]

```

In[ ]:= N[Log[E, 4]]

```

Out[]=

1.38629

```
In[*]:= N[Log[E, 2]]  
Out[*]= 0.693147
```

```
In[*]:= N[Pi]  
Out[*]= 3.14159
```